

UNIT 1. THE STRUCTURE OF EARTH

Primordial Earth formed around **4.6 billion years ago**. Due to cooling and rotation, substances in the gas, liquid and solid states separated and became arranged in spherical layers according to **density**. These arranged layers are called **geospheres**.

Geospheres are grouped into the outer and inner layers. The **outer** layers are the **atmosphere** - the layer of air, the **biosphere** - the layer of life, and the **hydrosphere** - the layer of water. The inner layers are the **crust**, the mantle and the **core**.

Definitions:

- **Hydrosphere:** the **non-contiguous** layer containing all the **waters** in any state on Earth. This includes **underground** water, locked or trapped in rocks, **surface** watercourse, lakes, seas, oceans and water vapour in the **atmosphere**.
- **Pedosphere:** the outermost, **non-contiguous** layer of the Earth's crust (on the weathering crust, in the soil, in the toposphere and in the hydrosphere). This soft and fertile layer provides **plants** with **water and nutrients**.
- **Biosphere:** the system containing all living beings within the **lithosphere**, lower **atmosphere** and **hydrosphere**.
- **Crust:** The **outermost** inner shell of Earth, it has the smallest mass and it is composed of **solid** rock. Average thickness is 30 km, the structure and thickness of **the oceanic** and **the continental crust** are different.
- **Mantle:** A **2900 km** thick layer located between the crust and the core. The upper mantle is composed of a layer of solid rock, with a layer of plastic, **molten rock** underneath, while the lower mantle is **solid**.
- **Core:** the hot and very dense innermost geosphere, about 700 km in a diameter, composed of **iron and nickel**. It is divided into two parts, a liquid **outer core** and solid **inner core**.
- **Geothermal gradient:** the rate of the inner temperature increase, on average, **3°C** every **100 metres**.

Questions

1. When did the primordial Earth form?
2. Which are the three outer geospheres?
3. What is the hydrosphere made of?
4. What happens to the temperature as we go deeper inside the Earth?